Krish Jeswal

AI/ML Engineer & Full-Stack Developer | Electronics & Telecom Undergrad
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Summary

Electronics & Telecom undergrad at RVCE (2028). I build AI/ML systems end to end - framing the evaluation, running the ablations, shipping the interface. I broke a masked AES-128 implementation with a profiling side-channel model and run an agentic RAG assistant on a locally fine-tuned Llama at zero infrastructure cost. I ship the product layer around them too, publishing TypeScript MCP packages and delivering a production React site as IETE Dev Lead.

Education

R.V. College of Engineering — Undergraduate
Electronics and Telecommunication — Bengaluru, India • 2024 - 2028

Amity International School — Higher Secondary
Physics, Chemistry, Maths and Computer Science — Ghaziabad, India • 2009 - 2023

Skills

Programming Languages
Python, TypeScript, JavaScript, C, SQL, HTML/CSS, Assembly, Verilog, GLSL, Bash

AI/ML & LLM Engineering
scikit-learn, PyTorch, RAG, Hugging Face, LangChain, QLoRa, SHAP, XGBoost, Ollama

Data, Retrieval & Storage
NumPy, pandas, MongoDB, PostgreSQL, Pydantic v2, ZenML, matplotlib/seaborn

Full-Stack Development
React, Vite, Three.js/WebGL, GSAP, Node.js, REST API, OAuth, MCP, Streamlit, Hono

Tools & Infrastructure
Docker, Firebase, Git, pytest, Ruff/oxlint, uv, Bun, Jupyter, Comet ML, Opik

Embedded Systems & Electronics
Arduino, STM32, ESP32, FPGA, LoRa, UART, RFID, ADC, LTSpice, KiCAD, Simulink

Projects

CipherTrace - ML Side-Channel Cryptanalysis — https://github.com/KrishJeswal/CipherTrace

- Built an end-to-end profiling side-channel attack pipeline against first-order Boolean-masked AES-128 (ASCAD, 50K ATMega8515 power traces), sweeping a 72-run grid of 6 classifiers × 3 POI strategies (SNR / ANOVA / PCA) × 4 feature budgets, evaluated by Guessing Entropy rather than accuracy.
- Achieved practical key recovery (GE = 0.46 at 500 attack traces, ~26× better than the Hamming-Weight baseline of 12.03) by switching from 9-class HW labels to 256-class Identity labels with an MLP over PCA components.
- Showed Macro F1 does not predict attack success (Decision Tree: best F1 0.1094, GE 148.86) and that ANOVA selection scores worse than random on masked traces; localized leakage via SHAP and shipped a Streamlit app for interactive attack configuration and GE analysis.

LoreRecall - Agentic Second Brain — https://github.com/KrishJeswal/LoreRecall

- Built five ZenML pipelines ingesting a Notion workspace via API, crawling every embedded link with crawl4ai, and filtering noise through a two-tier heuristic + LLM quality scorer; expanded ~100 source notes into ~600 clean documents at $0 infrastructure cost.
- Distilled a summarization dataset with Gemini 2.5 Flash and QLoRA fine-tuned Llama 3.2 3B (Unsloth, Kaggle T4), exporting to GGUF for local Ollama inference under 6 GB VRAM and eliminating per-chunk API cost on index rebuilds.
- Benchmarked three retrieval strategies (hybrid RRF, parent-document, contextual) over MongoDB Atlas vector + full-text search, serving the winner through a smolagents ToolCallingAgent with Opik tracing and LLM-as-judge evaluation

GlyphMark - Multi-Surface MCP Toolchain — https://github.com/KrishJeswal/GlyphMark

- Architected a Bun + TypeScript monorepo that exposes a single Telegram messaging core (Zod 4 schemas, typed operations) through four adapters - a Commander CLI, a stdio MCP server, a Hono-based remote MCP server, and an agent-facing Skill - with zero business logic outside the core package.
- Published three npm packages (@krish-dev/glyphmark-core, -cli, -mcp) built with tsdown to Node ESM, resolving workspace:* ranges at publish time and enforcing quality via oxlint, oxfmt, and strict tsc across the workspace.
- Secured the remote MCP surface with Clerk OAuth bearer verification and standards-based resource-metadata discovery, using stateless per-request server instances and a credential model that keeps Telegram bot tokens out of every MCP tool schema.

Experience

Google Developer Groups - Research & Projects — Jun 2025 – Present
Earned 19 Google Cloud Skill Badges and 1 Arcade through Google Cloud Study Jams, gaining hands-on experience with Google Cloud services, AI/ML, and application deployment; provided technical mentorship to 15+ teams at Vision 2047 Hackathon on ML pipeline architecture and cloud deployment; hosted a DevSprint session on open source contribution and cybersecurity fundamentals, guiding hands-on workflows and security best practices.

The Institution of Electronics and Telecommunication Engineers (IETE) - Development Lead — Nov 2025 – Present
Led the development team to architect and ship the official IETE Student Forum website - defined the architecture (React, TypeScript, Tailwind, PostgreSQL), established GitHub CI/CD Actions and drove end-to-end delivery from system design to production deployment. Introduced code review workflows and task delegation across the chapter.